MIRZA ABYAZ AWSAF

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ABOUT ME

I am an AI-Engineer with expertise in Python, FastAPI and PyQt, specializing in machine learning frameworks such as TensorFlow, PyTorch, and Keras. Proficient in Computer vision using OpenCV and skilled in data analysis with Scikit-learn, experienced in database management with SQL. I hold a strong passion for AI especially in Computer Vision and apart from AI I am really into Data Analysis, driving useful insights from data.

TECHNICAL SKILLS

Languages	Python, JAVA, C++,C, Javascript
AI / ML	TensorFlow, PyTorch, Keras, Scikit-learn,Langchain
Database	OracleDB, MariaDB, DBeaver
Desktop / Web	PyQt5, Qt Designer, Django,Node.js, React.js, FastAPI
Data Analytics	Power BI, Tableau, MS Excel
Tools	GitHub, Docker

WORK EXPERIENCE

AI Engineer | Regnum Resource Ltd. *2024 – Present*

- Designed and deployed the Automatic Vehicle Classifier (**AVC**) using YOLOv11 and RFDETR, classifying 13 vehicle types of **Road and Highways(RHD)** with a significant reduction in misclassification incidents at toll plazas.
- Built the Automatic Traffic and Vehicle Counting (**ATCC**) system incorporating real-time vehicle detection, tracking, speed estimation, lane-violation detection.
- Collaborated on the AI-powered Toll Management System(**TMS**) integrating **RFID**, **ANPR** and **AI** for toll plaza management, reducing manual intervention significantly.
- Delivered all systems as polished desktop applications using **PyQt5** and **QtDesigner** with real-time dashboards

EDUCATION

BSc, Computer Science and Engineering
BRAC University *2020 – 2025*

Advanced Level (A' Levels)
Bangladesh International Tutorial *2019*

Ordinary Level (O' Levels)
Bangladesh International Tutorial *2017*

Training & Certifications

Data Analyst Certification
Ostad *2025*

KEY PROJECTS

Automatic Vehicle Classifier (AVC)

Developed a real-time vehicle classification system using **YOLOv26** and **RFDETR** for accurate classification of 13 vehicle types, reducing vehicle misclassification incidents by 65% and improving toll classification integrity. Implemented an automated image logging pipeline organized by timestamp and vehicle class for monitoring and analysis. Designed and developed a desktop application using **Qt Designer** for the frontend and **PyQt5** for the backend.

Traffic Survey using Computer Vision (ATCC)

Developed an Automatic Traffic and Vehicle Counting (ATCC) system using computer vision and deep learning techniques for real-time traffic monitoring and analysis, it was built for the **Highways Department (HDM)**. Implemented vehicle detection, tracking, counting, lane violation detection, using **YOLOv11**, **YOLOv26**, **RTDTER**, **ByteTrack, OcSort**, **SAHI**, and other technologies. Designed the system to process live and recorded video streams with automated data logging for traffic analytics and reporting. Built a desktop-based monitoring interface using **PyQt5** to provide real-time visualization, statistics, and system control functionalities.

Toll Management System (TMS)

A toll operation software designed to automate and streamline toll plaza management using **RFID**, **ANPR**, and vehicle classification technologies for smooth and efficient transactions. Features real-time vehicle monitoring, automated logging, and intelligent verification to improve operational accuracy and reduce manual intervention. Built as a desktop application using **PyQt5** and **Qt Designer** and **OracleDB** for the database. It was built for **Roads and Highways Department (RHD)** of Bangladesh

DataForge: Local Dataset Management tool for Computer Vision tasks

This toolkit was developed to replace the dependency on the paid Roboflow version, it was designed as per its use case. It was built on **React** as its frontend and **Fast-AP**I for the backend. The tool allows to **analyze** the dataset, **extract** the desired data to mitigate the dataset imbalance issues. Moreover, it also allows the user to **generate** dataset using prebuilt models or just the raw data whichever is data and finally **annotate** the dataset locally without the risk of making it public.

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LANGUAGES

English (Fluent) **Bangla** (Native)